

SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 05

Life Processes (Full Chapter)

Plant & Human Nutrition • Photosynthesis • Human Digestion • Respiration Pathways • Circulatory System • Plant Transport • Human & Plant Excretion

Target Level: CBSE Class 10

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Chapter Overview & Basic Concepts

Living organisms perform basic maintenance activities continuously to stay alive and maintain cellular organization. The primary criterion to decide whether something is alive is **invisible molecular movement** within cells.

4 Essential Life Processes:

- **Nutrition:** Process of intake and utilization of nutrients by an organism.
- **Respiration:** Oxidative breakdown of food molecules inside cells to release ATP energy.
- **Transportation:** Internal movement of gases, nutrients, water, and metabolic wastes.
- **Excretion:** Removal of toxic metabolic nitrogenous wastes from the body.

NCERT High-Yield Insight: Simple diffusion is sufficient for single-celled organisms because their entire surface is in contact with the environment. In complex multicellular humans, diffusion is too slow (it would take ~3 years for oxygen to reach our toes by diffusion alone!).

SECTION 1: NUTRITION (PLANTS & HUMANS)

Autotrophic Nutrition	Heterotrophic Nutrition
Organisms synthesize organic food from simple inorganic raw materials (CO ₂ , H ₂ O) using sunlight and chlorophyll.	Organisms depend directly or indirectly on autotrophs for food.
Examples: Green plants, autotrophic bacteria.	Examples: Animals, fungi, humans, Amoeba.

Heterotrophic Nutrition Sub-types

- **Holozoic:** Ingestion, internal digestion, absorption, and assimilation of solid organic food (e.g., *Humans, Amoeba*).
- **Saprotrophic:** Organisms feed on dead/decaying matter by secreting enzymes outside (e.g., *Fungi, Yeast, Bread Mould*).
- **Parasitic:** Deriving nutrition from a host organism without killing it (e.g., *Cuscuta, Ticks, Lice, Tapeworm*).

1.1 Plant Nutrition (Photosynthesis)



3 Main Steps of Photosynthesis:

1. **Absorption** of light energy by chlorophyll.
2. **Conversion** of light energy to chemical energy and **Splitting (Photolysis)** of water ($2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$).
3. **Reduction** of Carbon Dioxide (CO₂) to Carbohydrates (Glucose).

Desert Plants Mechanism: Desert plants absorb CO₂ at night and store it as an intermediate compound. During daytime, chlorophyll absorbs solar energy to complete photosynthesis without keeping stomata open under intense heat.

Stomata & Guard Cells: Microscopic pores on leaf surfaces for gaseous exchange and transpiration. When water flows into guard cells, they swell (become turgid) and open the pore. When water flows out, they shrink (become flaccid) and close the pore.

1.2 Human Nutrition (Alimentary Canal & Enzymes)

Steps: Ingestion → Digestion → Absorption → Assimilation → Egestion.

- **Mouth (Salivary Amylase):** Converts Starch → Maltose.
- **Oesophagus:** Pushes food downward via **Peristalsis** (rhythmic muscular contraction/relaxation).
- **Stomach:**
 - **HCl:** Creates acidic medium (pH ~1.5–2.0) to activate Pepsin and kill bacteria.
 - **Pepsin:** Digests Proteins → Peptones.
 - **Mucus:** Protects inner stomach wall from HCl corrosion.
- **Liver:** Secretes Bile Juice (Emulsifies large fat globules into small droplets & provides alkaline medium).
- **Pancreas:**
 - **Pancreatic Amylase:** Breaks down Starch.
 - **Trypsin:** Converts Proteins → Peptides in alkaline medium.
 - **Lipase:** Converts Emulsified Fats → Fatty Acids + Glycerol.
- **Small Intestine:** Site of complete digestion. Contains finger-like projections called **Villi** that increase surface area for maximum absorption of nutrients into blood.
- **Large Intestine:** Reabsorbs excess water and salts from undigested waste.

SECTION 2: RESPIRATION (PLANTS & HUMANS)

Respiration is the cellular process of breaking down glucose to release energy stored as **ATP** (1 mole ATP = 30.5 kJ/mol).

Glucose Breakdown Pathways (NCERT Core Chart)

Glucose (6-Carbon) —[In Cytoplasm: Glycolysis]→ Pyruvate (3-Carbon) + Energy

- **Absence of O₂ (Yeast / Fermentation):** Ethanol (2C) + CO₂ + 2 ATP.
- **Lack of O₂ (Human Muscle during heavy exercise):** Lactic Acid (3C) + 2 ATP (*Causes muscle cramps*).
- **Presence of O₂ (Mitochondria):** CO₂ + H₂O + 38 ATP (*Aerobic Respiration*).

2.1 Human Respiratory System

Pathway: Nostrils → Nasal Passage → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli.

- **Rings of Cartilage:** Present in trachea to prevent it from collapsing when air pressure drops.
- **Alveoli:** Thin-walled balloon-like structures wrapped with blood capillaries. Provides ~80 m² surface area for gas exchange.
- **Hemoglobin:** Respiratory pigment in RBCs that carries O₂. CO₂ is mostly transported dissolved in blood plasma.

2.2 Plant Respiration

Plants exchange gases via Stomata (leaves), Lenticels (stems), and Root surfaces.

- **Daytime:** Photosynthesis rate > Respiration rate → Net release of O₂.
- **Nighttime:** Photosynthesis stops → Net elimination of CO₂.

SECTION 3: TRANSPORTATION (PLANTS & HUMANS)

3.1 Human Circulatory System

Consists of Heart, Blood Vessels, and Blood/Lymph.

Double Circulation: Blood passes through the heart twice in one complete cycle:

1. **Pulmonary Circuit:** Right Ventricle → Pulmonary Artery → Lungs → Pulmonary Veins → Left Atrium.
2. **Systemic Circuit:** Left Ventricle → Aorta → Body Parts → Vena Cava → Right Atrium.

Feature	Arteries	Veins	Capillaries
Function	Carry blood <i>away</i> from heart	Carry blood <i>towards</i> heart	Exchange materials between blood and cells
Pressure & Wall	High pressure; thick, elastic walls	Low pressure; thin walls	Low pressure; single-cell thin wall
Valves	Absent	Present (prevents backflow)	Absent

Platelets & Lymph: Platelets plug leaks by forming blood clots. Lymph is a colorless tissue fluid without RBCs that drains excess fluid back into veins and carries digested fats from intestine.

3.2 Plant Transport System

Feature	Xylem	Phloem
Transport Material	Water and Minerals	Photosynthetic Products (Sucrose, Amino Acids)
Direction	Unidirectional (Roots → Leaves)	Bidirectional (Source ↔ Sink)
Mechanism	Root Pressure (night) & Transpiration Pull (day)	Translocation using energy (ATP)

SECTION 4: EXCRETION (PLANTS & HUMANS)

4.1 Human Excretory System & Nephron

Organs: Pair of Kidneys, Pair of Ureters, Urinary Bladder, Urethra.

Nephron Mechanics: Each kidney contains ~1 million nephrons.

1. **Ultrafiltration:** Blood is filtered under high pressure in the Glomerulus inside Bowman's Capsule.
2. **Selective Reabsorption:** Useful substances like glucose, amino acids, salts, and water are reabsorbed in the renal tubule back into capillaries.
3. **Urine Secretion:** Concentrated urine containing urea, uric acid, and excess salts enters Collecting Duct → Ureter → Bladder → Urethra.

4.2 Plant Excretion

- **Gaseous Wastes (O_2 , CO_2):** Removed via stomata and lenticels.
- **Water:** Excreted via Transpiration.
- **Other Wastes:** Stored in cellular vacuoles, old leaves that fall off, or deposited as Resins and Gums in old xylem.

? Important Practice Questions

Q1: What is the role of HCl in the stomach?

Ans: Creates an acidic medium (pH ~1.5–2.0) to activate Pepsin and kills harmful ingested bacteria.

Q2: Why do muscle cramps occur during heavy exercise?

Ans: Due to lack of oxygen in muscle cells, pyruvate undergoes anaerobic respiration to form Lactic Acid, which accumulates and causes muscle cramps.

Q3: How are alveoli designed to maximize gas exchange?

Ans: They have balloon-like structures with thin single-celled walls, an extensive capillary network, and a large combined surface area (~80 m²).

Q4: Differentiate between Xylem and Phloem transport.

Ans: Xylem transports water/minerals unidirectionally using physical suction (transpiration pull). Phloem translocates sucrose bidirectionally using active transport (ATP energy).